

Valuation of Auto-Callable Barrier Notes Linked to Applied Materials via Binomial Trees

1 Summary

In this project, we price a Contingent Coupon Auto-Callable Notes linked to the common stock of Applied Materials, Inc. (AMAT) by binomial tree.

By constructing this binomial tree which captures auto callable feature and contingent coupon, the estimated value of the Notes is between \$9.575 and \$9.755 per \$10 principal amount Note.

2 Data

We download interest rate, dividend yield, and implied volatility from Bloomberg. From the file, this product has trade date on January 24, 2025, issue date on January 29, 2025, valuation date on January 23, 2026 and maturity date on January 28, 2026.

Using SWDF command in Bloomberg, we extract discount rate starting on January 24, 2025 (trade date) from OIS term structure. Note that following market convention, the interest rates are quoted on January 22, 2025. Using linear interpolation, we get the discount rate through January 23, 2026 (valuation date) is 0.958940, the interest rate converted from this value stands for expected return of the underlying asset, through January 28, 2026 (maturity date) is 0.958414, the interest rate converted from this value stands for time value of money.

	Oct. 24, 2025	Jan. 26, 2026	July 24, 2026
Discount rate	0.968590	0.958622	0.939981

Table 1: Discount rate of OIS

Using OVME command, we get dividend yield of AMAT is 1.011% between January 24, 2025 (trade date) and January 23, 2026 (valuation date), which is actually calculated by method from OPDF command.

Using OVM command, we get volatility matrix of AMAT, we interpolate the potential volatility on the observation dates that can be used to construct the binary tree as follows:

Moneyiness	April 23, 2025	July 23, 2025	Oct. 23, 2025	Jan. 23, 2026
65%	0.47029	0.42916	0.41531	0.41494
100%	0.37618	0.37465	0.37894	0.37798

Table 2: Volatility Matrix of AMAT

3 Algorithm

The product has two main features, automatic call and contingent coupon. To be specific, for each quarterly Observation Date, if the underlying close price is equal to or greater than its initial value, the Notes will be automatically called and pay the principal and contingent coupon, then no further payments will be made on the Notes, if the underlying close price is equal to or greater than the coupon barrier, the Notes will pay a contingent coupon on coupon payment date, if the underlying close price is less than coupon barrier, the Notes will not make any payment.

We construct a binomial tree adjusting for these two features on observation dates as follows.

(1) Parameters of tree,

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad q = \frac{e^{(r_1 - \delta)\Delta t} - d}{u - d}$$

where σ is the volatility of AMAT, Δt is the time step, r_1 is the expected return between trade date and valuation date, and δ is the dividend yield.

(2) On valuation date,

$$V_{N,j} = \begin{cases} (10 + 0.2838) \times e^{-r_2(Nc-N)\Delta t}, & \text{if } S_{N,j} \geq \text{downside threshold,} \\ 10 \times (1 + \text{underlying return}) \times e^{-r_2(Nc-N)\Delta t}, & \text{else.} \end{cases}$$

where Nc is the number of time steps of maturity date, N is the number of time steps of valuation date, j is the number of up ticks, principle = \$10, contingent coupon = \$0.2838,

$$\text{underlying return} = \frac{\text{Final value} - \text{Initial value}}{\text{Initial value}}.$$

where the final value is determined on valuation date, initial value S_0 is \$190.70, the close price on Jan. 23, 2025. Downside threshold is $65\% \times S_0 = \$123.98$. r_2 is the risk-free rate discounting time value between trade date and maturity date.

(3) On any observation date,

$$V_{i,j} = \begin{cases} (10 + 0.2838) \cdot e^{-r_2(ic-i)\Delta t}, & \text{if } S_{i,j} \geq S_0, \\ e^{-r_2\Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}] + 0.2838 \cdot e^{-r_2(ic-i)\Delta t}, & \text{if coupon barrier} \leq S_{i,j} < S_0, \\ e^{-r_2\Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}], & \text{else.} \end{cases}$$

where i is the number of time steps of observation date, ic is the number of time steps of the first coupon payment date following the applicable observation date. Coupon barrier is $65\% \times S_0 = \$123.98$.

(4) On any other date,

$$V_{i,j} = e^{-r_2 \Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}]$$

where i is the number of time steps.

4 Discussion

The binary tree model is easy to understand, but it also requires many assumptions that can affect the accuracy of the pricing.

1. **Dividends:** Individual stock generally pays discrete dividends. If we consider this scheme, the adjustment of valuation tree and estimation of dividends amount are complicated, hence we just assume a continuous dividend yield for simplicity in our pricing process.
2. **Interest rates:** In binomial tree, there are multiple cash flows at different time steps, and interest rates are actually inconsistent in each time step, so a more theoretically correct way is to use different forward rates at different periods. However, as we know, equity derivatives are somewhat less sensitive to interest rates than other more important factors (e.g. volatility) or other products (e.g. bonds), we still use long term interest rates to compute stock prices and discount factors for simplicity. We use interest rate between trade date and valuation date for valuation and interest rate between trade date and maturity date for discounting. Note that it is also reasonable to use forward rate between issue date and maturity date for discounting, but for the reasons mentioned above, we didn't try too much.
3. **Volatility:** In CRR model, up and down ticks are determined by volatility. Volatilities vary with strikes and maturities in real market, yet we assume a constant volatility in the tree. On one hand, though we can regard volatility as a function of time and calculate a forward volatility like a forward interest rate, different volatilities at different times may cause the failure of recombination of the tree and increase computational complexity in standard model. Some papers also show that building a fully flexible tree (Derman-Kani tree) also doesn't show better results than simple models. On the other hand, we can't explicitly specify using different strikes in the same tree at the same time, and the effect of strike to volatility is larger than the effect of time, so our main concern turns to choose a proper constant strike for consistent volatility in the whole tree. If the volatility is too small, the probability of hitting autocall condition or coupon barrier will be low, resulting in an overvaluation, and if the volatility is too big, the probability of reaching a bad state will be high, resulting in an

Choice	Volatility	Price
ATM volatility at maturity	0.37798	9.752
mean of ATM volatility on observation dates	0.37694	9.755
coupon barrier volatility at maturity	0.41494	9.608
mean of coupon barrier on observation dates	0.43242	9.575

Table 3: Volatility and Price Comparison

undervaluation. We tried several common choices and listed the results below.

4. **Time steps:** all above results are calculated with a daily time step (1/365), if we change the step size, the estimated price will also change due to the distribution error. For example, when it is 1/365/2, the prices will be:

Choice	Price
ATM volatility at maturity	9.789
mean of ATM volatility on observation dates	9.791
coupon barrier volatility at maturity	9.677
mean of coupon barrier on observation dates	9.569

Table 4: Option price when scale = 2

From Figure 1 : Effect of scale on pricing, we can see as the scale increases, the price calculation becomes more refined, with smaller time steps leading to more precise valuations. At lower scales, the price is slightly higher due to a coarser binomial tree, which may overestimate the impact of upward movements. As the scale increases, the price gradually adjusts, reflecting a more accurate estimate of the structured note's value. The true price lies within this range, meaning that choosing an appropriate scale is essential for balancing computational efficiency and accuracy.

5. **Nonlinearity error:** the payoff functions of this product are not sufficiently smoothed because of the autocall and coupon features, which will lead to some nonlinearity error.

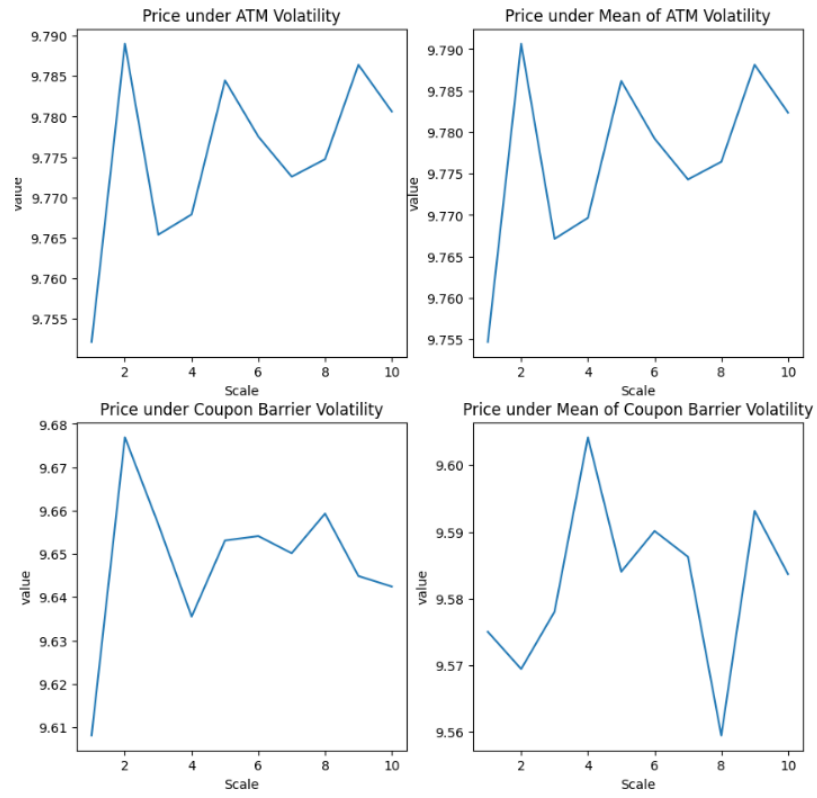


Figure 1: Effect of Scale on Pricing

Appendix

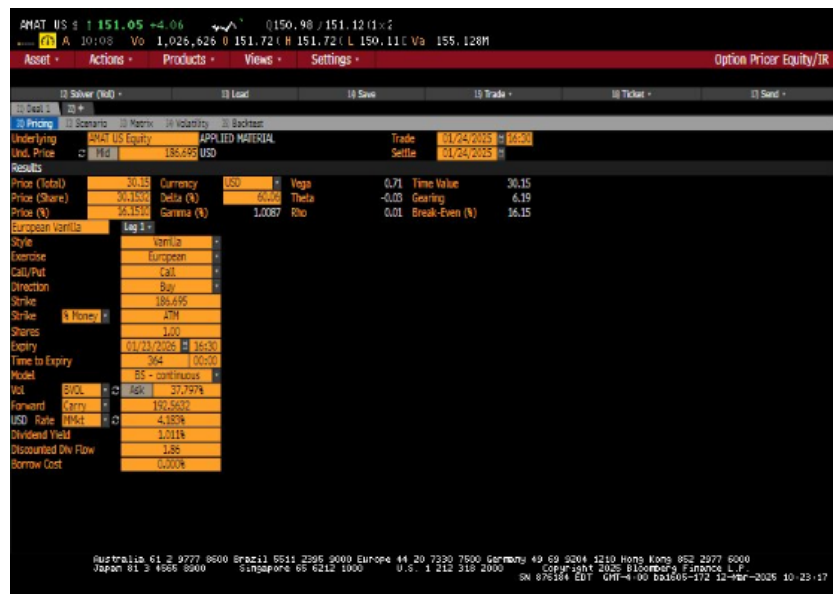
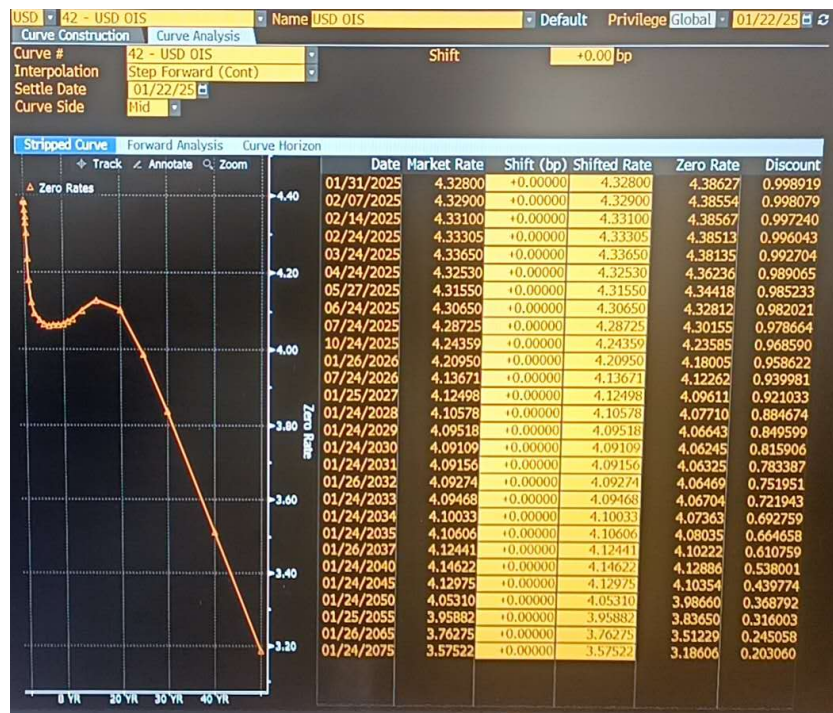




Figure 4: Implied Volatility